

Qatar-1b

R-1570

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_211011 · created 2026-08-01T22:30:58+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459498.7416 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.012
Transit depth (Rp/Rs)^2	3.43 +/- 0.43 [%]
Orbital Inclination (inc)	83.09 +/- 0.78
Airmass coefficient 1 (a1)	1.179 +/- 0.017
Airmass coefficient 2 (a2)	-0.111 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.43 %
Best Comparison Star	#1 - [477, 99]
Optimal Method	PSF photometry
Transit Duration (day)	0.0664 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.185 ± 0.012 vs 0.1463 (deviation 3.22σ)
Duration fit vs published	0.0664 d vs 0.0692 d (deviation 0.51σ)
Mid-time O–C	3.3 min
Depth SNR	15.4
Residual scatter	1.43 %

Light curve

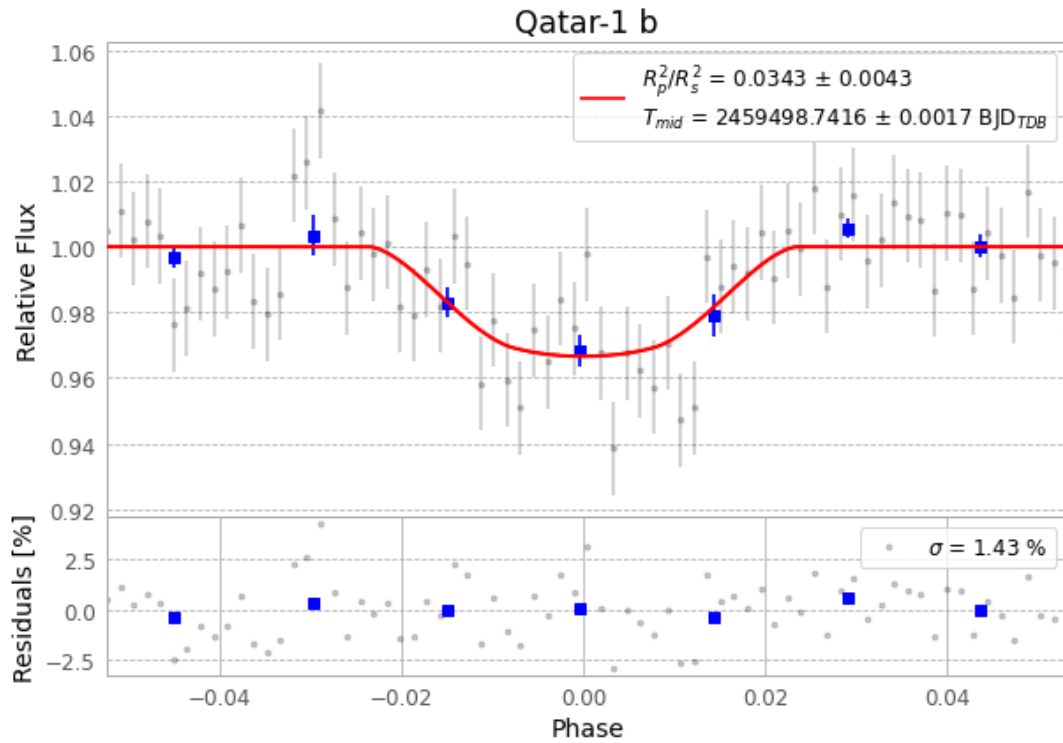


Figure 1 — FinalLightCurve_Qatar-1 b_2021-10-10.png (SHA-256 60128f3825cf45d6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [483, 292], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b1498598e117f70b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), Qatar-1211011035712.FITS → Qatar-1211011073312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-211011.log (7c3820c46b45...), FinalLightCurve_Qatar-1 b_2021-10-10.png (60128f3825cf...), FinalLightCurve_Qatar-1 b_2021-10-10.csv (dffff6a5063e...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 22:30Z.